Why is Amstelland's water still not clean, and what can Waternet do about it?

A data and AI study of 45 years of water measurements by Waternet, the water authority for Amsterdam and the surrounding area. Written so that anyone can follow it, no water expertise needed.

How to read this page

1. The question: what we wanted to find out and why it matters
2. The data: what we used, and how we cleaned it
3. What the data shows: trends, problem spots, and which problems occur together
4. How we analysed it: our choices and the models, explained
5. Results: why algae grows where it does, whether Waternet's fix works, early warnings, climate
6. Conclusions and suggested actions

1 · The question

The problem in plain words. Too many *nutrients* (phosphorus and nitrogen, from farmland, peat soils and sewage) act like fertiliser in the water. They feed *algae*. Too much algae makes the water green and murky, uses up oxygen so fish die, and can make swimming unsafe (some blue-green algae are toxic).

Our question: the water got much cleaner in the 1980s–90s, but has stopped improving. *Why do some places stay polluted, and where should Waternet act first?*

Why it matters: Waternet's own research plan asks for exactly this: a better understanding of water quality "through automated analysis and open data", and better phosphorus-removal plants.

2 · The data

Dataset	What it is	Size	Used for
Water quality lab measurements Waternet	Water samples taken across the area and tested in a lab: nutrients, algae, oxygen, clarity, bacteria, salt, etc.	2.1 million measurements · 1,321 sampling spots · 1963–2025	Everything below
Pumping stations Waternet	How much water each polder pump moves, every hour	83 pumps at 44 stations · 1996–2025	Water flow on maps; "flushing" as a possible cause
Pumping station locations Waternet public register (online)	Where each pump is and which way it pumps (not in the provided data, so we looked it up)	39 of 44 stations found	Placing pumps on the map
Weather KNMI, station Schiphol (online)	Daily rain, temperature, sunshine, wind	1990–2025	Weather as a possible cause
Ecology Waternet	Counts of plants, fish and small water animals	1.3 million observations	Not used: rarely measured at the same place and time as the chemistry
Pesticides Waternet (part of the lab data)	492 different pesticides	85,000 measurements, but only 2–4 spots per year since 2012	Checked, not used: too little recent data, and 85% below the detection limit

Cleaning choices

1. **Removed measurements Waternet flagged as rejected** (about 1% of all data).
2. **Removed 12 sampling spots that are not open water**: points inside a heat-exchange installation and the treated outflow of two plants.
3. **Values "below detection limit"** (too small for the lab to measure) were set to half that limit, the standard approach.
4. **Summer only (April–September)** for maps and trends: that's when algae grow and when water quality is officially judged.
5. **Typical value = median** (the middle value), not the average, so a few extreme samples don't distort the picture.

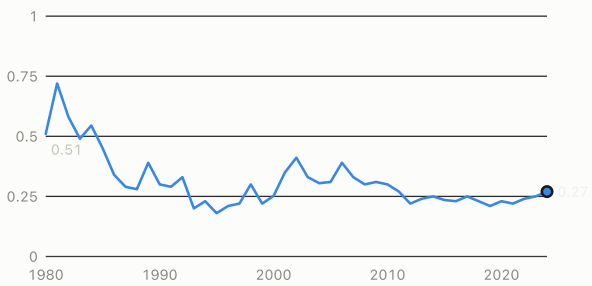
3 · What the data shows

3a · Trends: big improvement, then a standstill

Typical summer value across all sampling spots, per year. Hover a chart to read a year. Each chart has its own scale.

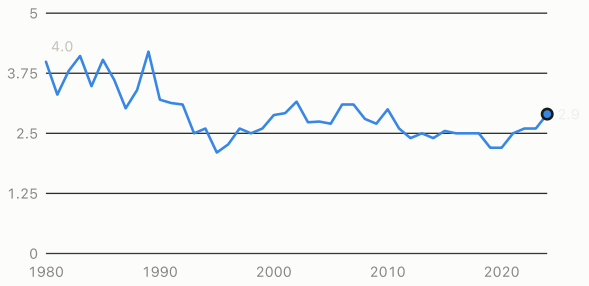
Phosphorus (mg/l)

Main nutrient feeding algae · lower is better



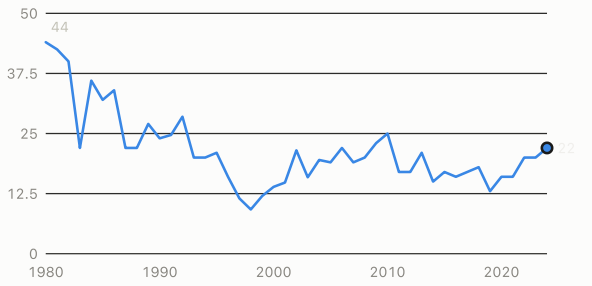
Nitrogen (mg/l)

Second nutrient, from fertiliser & sewage · lower is better



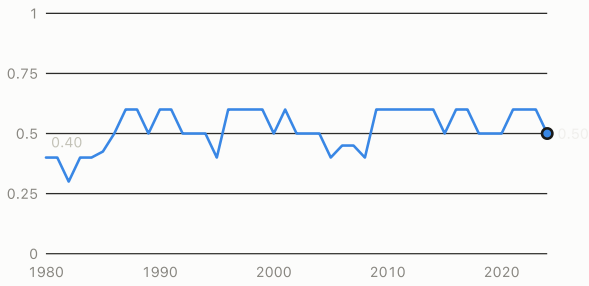
Algae: chlorophyll-a (µg/l)

How much algae is in the water · lower is better



Clarity (m)

How deep you can see · higher is better



Before ~1989 fewer than 50 spots were sampled (about 200 today), so early years are less representative.

▶ Show as table

> → **Finding:** phosphorus roughly halved and algae dropped by more than half between 1980 and 2000. Since about 2010 nothing has improved, and the water is still murky (you can see only ~0.5 m deep).

3b · Where are the limits broken? (2020–2024)

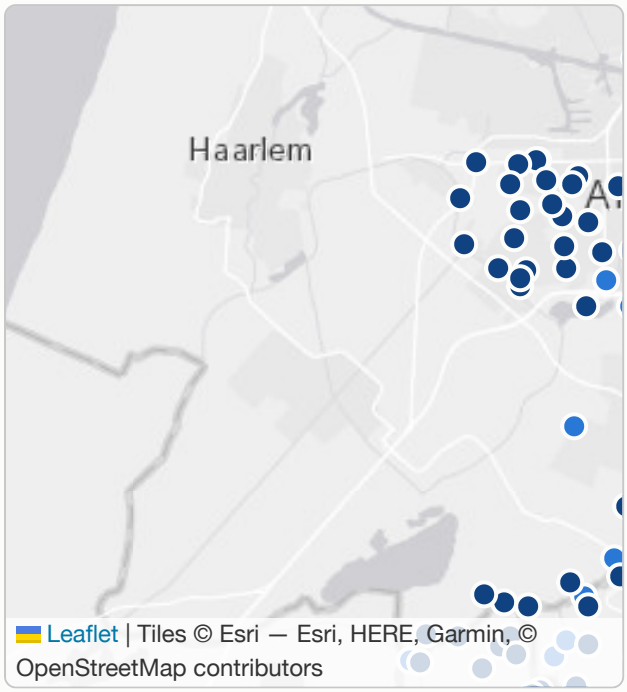
Each map shows one measure. Only spots that break the limit are colored (typical summer value 2020–2024); small gray dots are spots within the limit. Pan or zoom one map and the others follow. Limits are guide values close to typical Dutch water-quality targets; official targets differ per type of water.

Phosphorus: 176 of 241 spots above 0.15 mg/l (73%)

Nutrient that feeds algae

Above limit (70) Above 0.3 mg/l (106)

Within limit (65)

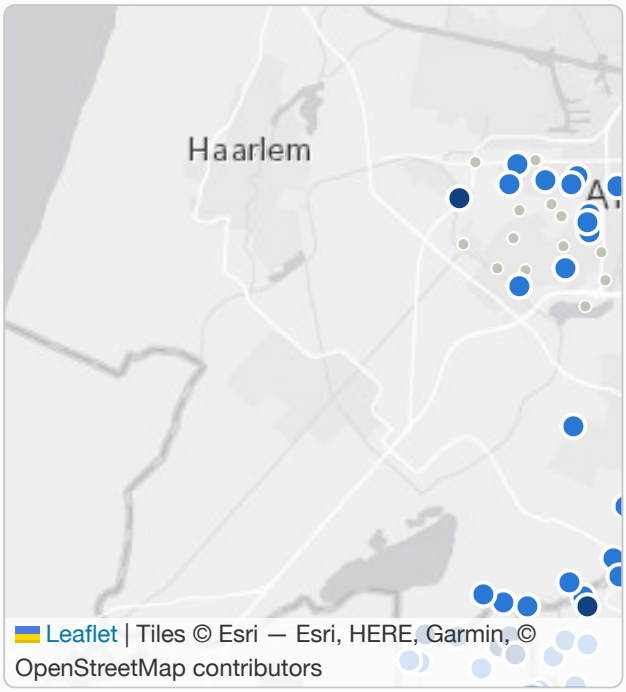


Nitrogen: 128 of 239 spots above 2.4 mg/l (54%)

Nutrient from fertiliser & sewage

Above limit (103) Above 4.8 mg/l (25)

Within limit (111)

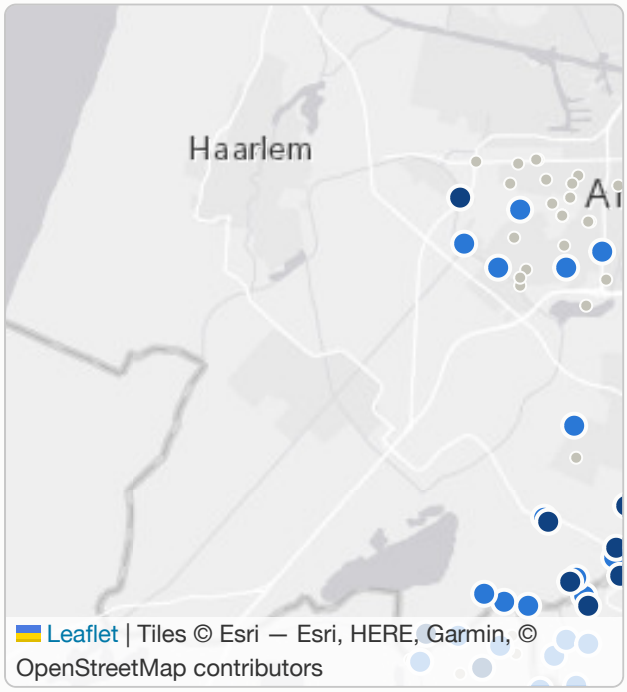


Algae (chlorophyll-a): 102 of 259 spots above 25 µg/l (39%)

How much algae is growing

Above limit (77) Above 100 µg/l (bloom) (25)

Within limit (157)

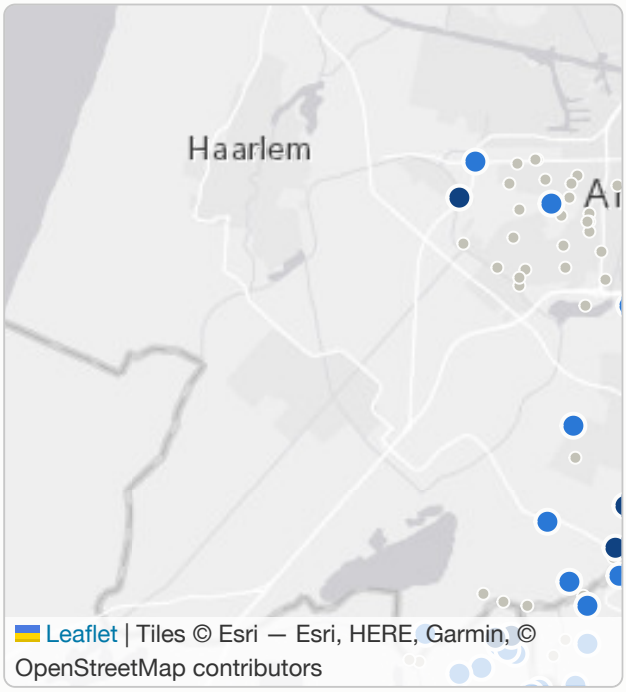


Clarity: 81 of 255 spots below 0.4 m (32%)

How deep you can see

Below limit (70) Below 0.2 m (11)

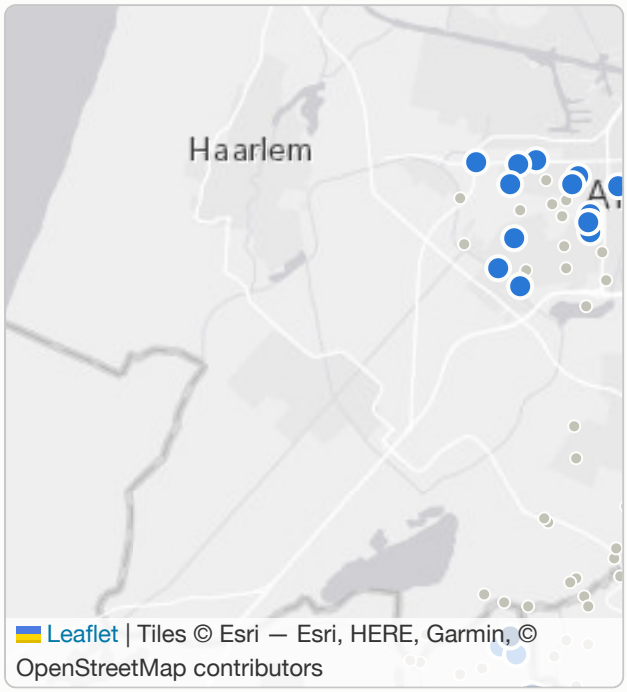
Within limit (174)



Oxygen: 57 of 266 spots below 5 mg/l (21%)

Fish need it

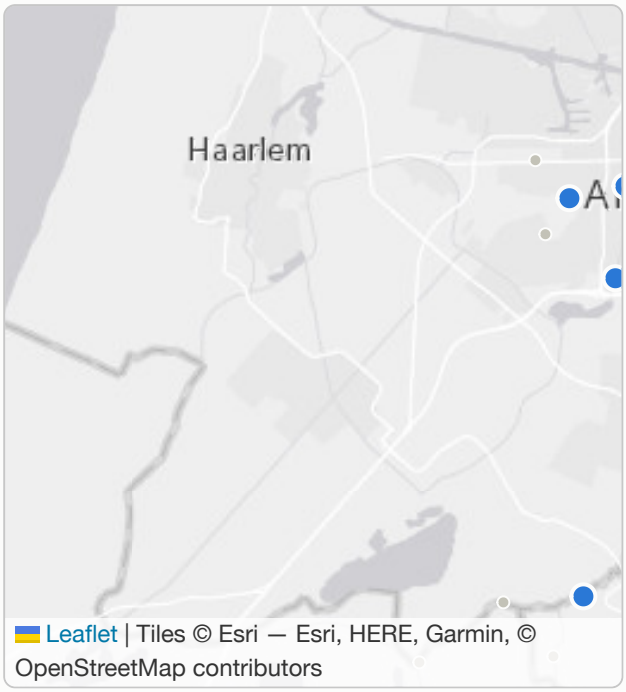
Below limit (51) · Below 3 mg/l (6) · Within limit (209)



E. coli bacteria: 19 of 60 spots above 200 per 100 ml (32%)

Sign of sewage · measured at swim spots only

Above limit (15) · Above 500 per 100 ml (4) · Within limit (41)



Dots: sampling spots with at least 5 summer measurements in 2020–2024. Arrows: average pumped volume 2020–2024 for 39 pumping stations, located with Waternet's public pump register (5 more stations in the data have no published location: KGM00153, KGM00518, KGM00230, KGM00122, KGM00362). 2,500 m³ is about one Olympic swimming pool.

3c · Connecting the dots: which problems occur together, and where?

We combined the five main measures (phosphorus, nitrogen, algae, clarity, oxygen) for every spot that has at least four of them, and counted how many limits each spot breaks.

Spots compared

235

with at least 4 of the 5 measures

Break 3 or more limits

110 (47%)

several problems at once

Break no limit

52 (22%)

clean on all measured counts

Number of limits broken per spot

Darker = more problems at once

- 0 limits (52)
- 1 limit (35)
- 2 limits (38)
- 3 limits (58)
- 4 limits (46)
- 5 limits (6)



By type of area: share of spots breaking each limit

Darker cell = larger share. This shows which kinds of places have which problems.

Type of area	phosphorus	nitrogen	algae	murky water	low oxygen	3+ limits
Peat ditches (85)	84%	66%	51%	36%	20%	**60%**
Clay ditches (32)	88%	63%	50%	44%	41%	**66%**
City canals & ditches (60)	87%	57%	29%	20%	35%	**47%**
Lakes (44)	23%	11%	11%	16%	7%	**5%**

The most common combinations of problems, and what they mean

Each row is a pattern of broken limits, with how many spots show it

Limits broken	Spots	Most common area	What it likely means
none	52	Lakes (58%)	Clean on all measures.
phosphorus + nitrogen + algae + murky water	33	Peat ditches (61%)	Classic over-fertilised water: nutrients feed algae, which make the water murky.
phosphorus + nitrogen + algae	32	Peat ditches (50%)	Nutrients and algae are both high: over-fertilised water.
phosphorus	29	City canals & ditches (31%)	Nutrients are high but algae stays low: something else holds algae back (shade, flow, duckweed or plants). Risky if that changes.
phosphorus + nitrogen + low oxygen	20	City canals & ditches (55%)	Low oxygen without much algae: likely rotting material (peat, mud, sewage) using up oxygen.
phosphorus + nitrogen	12	City canals & ditches (50%)	Nutrients are high but algae stays low: something else holds algae back (shade, flow, duckweed or plants). Risky if that changes.
phosphorus + nitrogen + murky water + low oxygen	11	Peat ditches (45%)	Low oxygen without much algae: likely rotting material (peat, mud, sewage) using up oxygen.

→ **Finding:** problems don't come alone. 110 of 235 spots (47%) break 3 or more limits at once. Clay ditches have the most stacked problems: 66% break 3 or more limits, versus 5% for lakes.

4 · How we analysed it

Our main choices, and why

1. **Algae is the main outcome** we explain (measured as chlorophyll-a, the green pigment in algae). It's the visible result of nutrient pollution.
2. **No "cheating" inputs.** We left out things that algae themselves change, such as clarity, oxygen and total nitrogen and phosphorus (which include the nutrients *inside* the algae). Otherwise a model could "predict" algae by looking at its effects.
3. **Strict testing.** Every model is tested on sampling spots and/or years it never saw during training, and compared to a simple guess. A model only counts if it clearly beats the simple guess.
4. **Each spot is linked to its nearest pumping station** (within 3 km), and the whole area to one weather station. Both are simplifications.
5. **We checked our own results for stability**, and corrected two problems we found along the way (an unstable setting, and nutrients behaving "backwards", see Model 1).

The models and analyses

Both models use **gradient boosting**: a standard machine-learning method that combines hundreds of small decision rules ("if the soil is peat and it was sunny, expect more algae"), each one correcting the mistakes of the previous ones. It works well on tables of measurements like ours.

Model 1 · Algae explainer

Gradient boosting (regression)

Question	How much algae do we expect in a water sample, given its conditions? And which cause matters most at each spot?
Inputs	Dissolved nutrients in the sample; type of place (soil, seepage, salt, sulphate, ditch or lake); nearest pump's flow; weather in the days before; time of year
Output	Expected algae (µg/l), then per spot the main driver via "what if this cause were typical?"
Trained on	47,476 algae samples, 2000–2025
Tested how	On sampling spots it never saw, and on future years (trained to 2019, tested 2020–2025)
Result	Explains **44%** of the ups and downs in algae at unseen spots (time of year alone: 9%). 53% of estimates within a factor of 2. Ranks dirty vs clean spots well (correlation 0.69).
In plain words	It has learned how algae responds to its causes, well enough to explain places it has never seen.
Limits	Explains, doesn't forecast. Underestimates extreme blooms. Dissolved nutrients partly reflect algae using them up.

Model 2 · Bloom forecaster

Gradient boosting (regression + classification)

Question	How much algae will a spot have at its next sample, and will it be a bloom (more than 100 µg/l)?
Inputs	Only what's known on the day of the last sample: that sample, the spot's bloom history, type of place, planned date
Output	Expected algae at the next sample (~4 weeks later) and a bloom probability
Trained on	30,652 pairs of consecutive samples, 2000–2019
Tested how	On 11,001 forecasts in 2020–2025 (never seen), against "same as last time"
Result	Catches **527 of 997** blooms vs 453 for "same as last time" (same number of warnings). R^2 0.55 vs 0.24.
In plain words	A month ahead, it points to the right places more often than simply assuming nothing changes.
Limits	Still misses about half of blooms; about half of warnings are false. Limited by lab samples ~4 weeks apart.

Analysis 3 · Plant comparison

Paired before/after measurement (no AI)

Question	How much phosphorus do Waternet's two phosphorus-removal plants remove?
Method	Water going in vs coming out on the same day; median removal with a 95% range (bootstrap) and a statistical test
Result	Botshol **71%** (131 days), Vinkeveen **30%** (143 days), also at equal inflow
Limits	Different sample types at the two plants; measures the plant, not yet the lake

Analysis 4 · Climate stress test

What-if scenario with Model 1

Question	How much more algae if summers are 1–2 °C warmer?
Method	Re-run Model 1 on recent summer samples with higher air and water temperature, all else equal
Result	+4% algae at +1 °C, **+7%** at +2 °C
Limits	Likely an underestimate (the model never saw water above 24 °C); a pattern-based what-if, not a guarantee

5 · Results

5a · Why does algae grow where it does? (Model 1)

For each spot we ask the model: *"what if this cause were at a typical level?"* The cause that would remove the most algae is that spot's **main driver**.

Variation in algae explained

44%

at spots the model never saw · season alone: 9%

Predictions within a factor of 2

53%

season alone: 35%

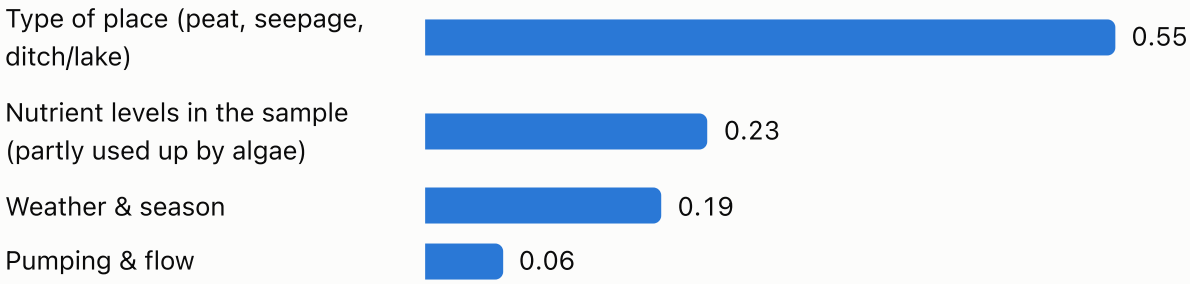
Spots explained

259

with ≥5 summer samples 2020–2024

What drives algae across the whole area?

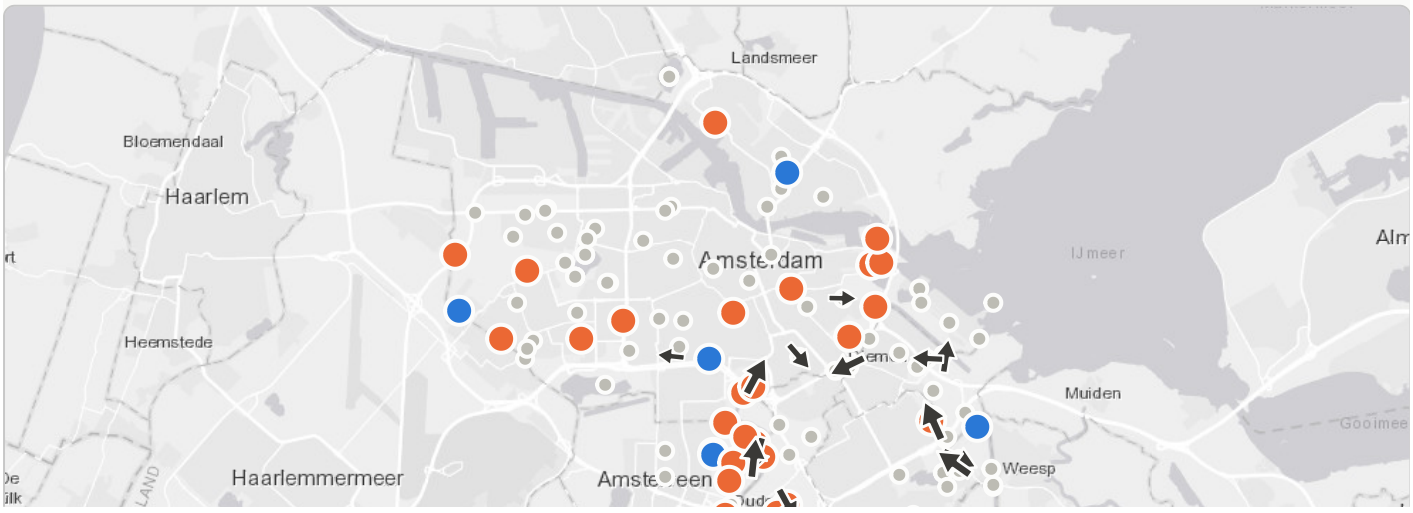
How much worse the model gets when it can't use a group of causes (higher = more important)

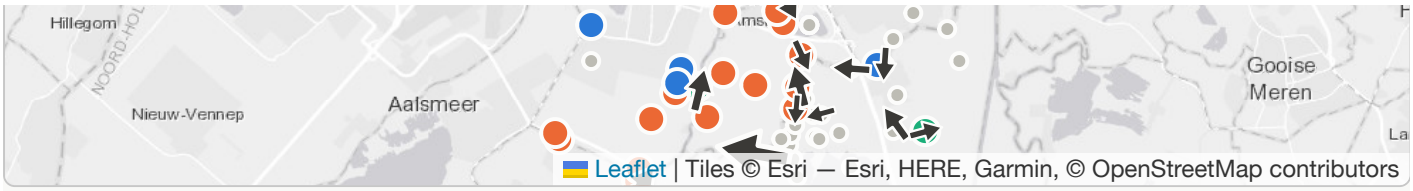


Main driver per algae hotspot (summers 2020–2024)

Nutrient levels in the sample (partly used up by algae) (21 spots)

Type of place (peat, seepage, ditch/lake) (74 spots) Pumping & flow (7 spots) Little algae (157 spots)





Colored: spots with a lot of algae (more than 25 µg/l), by main driver. Gray: little algae. Hover a dot for details.

Top 15 algae hotspots and what drives them

Sampling spot	Algae (µg/l)	Main driver	× more algae	Soil
hoge kant stuw, uitgang moerasblok 3 (Zuid) bij kruizing Midden- en Hoofdtocht, hoofdweg Vinkeveen	225	Type of place (peat, seepage, ditch/lake)	3.3×	Zware klei
hoge kant stuw bankrasweg Amstelveen repr voor nat gebied Tulpenburg	210	Nutrient levels in the sample (partly used up by algae)	2.4×	Veen
Voor krooshek gemaal aan het eind Raasdorperweg	210	Type of place (peat, seepage, ditch/lake)	4.5×	Veen / Zware klei
Voor krooshek gemaal Holendrechter en Bullewijker langs de Amstelweg	205	Type of place (peat, seepage, ditch/lake)	2.9×	Veen
langs Meijslootpad nabij golfterrein Amstelveen	190	Nutrient levels in the sample (partly used up by algae)	2.2×	Zware klei
Voor krooshek gemaal Bijlmer Zandpad langs Provinciale weg	185	Type of place (peat, seepage, ditch/lake)	3.6×	Bebouwing
mp vegetatie	183	Type of place (peat, seepage, ditch/lake)	1.7×	Veen
mp macrofauna, mp vegetatie	165	Type of place (peat, seepage, ditch/lake)	4.0×	Veen / Zware klei
Voor krooshek gemaal Bovenkerkerpolder	160	Pumping & flow	2.5×	Veen
Oosttocht brug bij Middelhoeve (nummer 12A), mp macrofauna, mp vegetatie	155	Type of place (peat, seepage, ditch/lake)	2.4×	Zware klei
Mennonietenwetering zuid van stuw	150	Type of place (peat, seepage, ditch/lake)	2.4×	Veen
hoge kant stuw bovenkerker molentocht instroom gemaal	135	Nutrient levels in the sample (partly used up by algae)	2.2×	Zware klei
Voor krooshek gemaal Rondehoep langs Waver	135	Type of place (peat, seepage, ditch/lake)	2.7×	Veen
Voor krooshek gemaal Eerste Bedijking	130	Type of place (peat, seepage, ditch/lake)	2.2×	Veen
langs mijdrechtse dwarsweg benadering einde van aan te leggen moerasblok	130	Type of place (peat, seepage, ditch/lake)	2.7×	Veen

"× more algae" = how much higher the model expects algae to be here because of that cause, compared to the same spot with typical levels of it.
Type of place = peat or clay soil, seepage (salt, sulphate), brown peat water, ditch or lake. It also captures nutrients that peat soils and ditch bottoms keep releasing. It points to a pattern; it doesn't prove the mechanism.

Nutrient levels in the sample: treat with care. Algae use these nutrients up, so a sample with a lot of algae often shows *low* dissolved nutrients. The model partly learned that effect, so this label is less reliable.

> → **Finding:** at **74 of 102** algae hotspots the main driver is the *type of place* (peat, seepage, ditch bottoms), not the nutrients flowing in today. Cutting nutrient inflow alone won't fix those spots.

5b · Does Waternet's own fix work? Phosphorus-removal plants

Waternet runs plants that strip phosphorus from water before it flows into the Botshol and Vinkeveen lakes. We compared water going **in** and coming **out**, sampled on the same day. This is a direct measurement, no model needed.

Botshol: phosphorus removed

71%

95% range 67%–74% · 131 same-day pairs · in 0.21 → out 0.06 mg/l

Vinkeveen: phosphorus removed

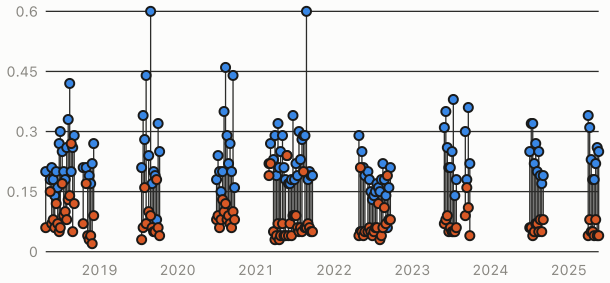
30%

95% range 20%–33% · 143 same-day pairs · in 0.17 → out 0.13 mg/l

Phosphorus going in Phosphorus coming out

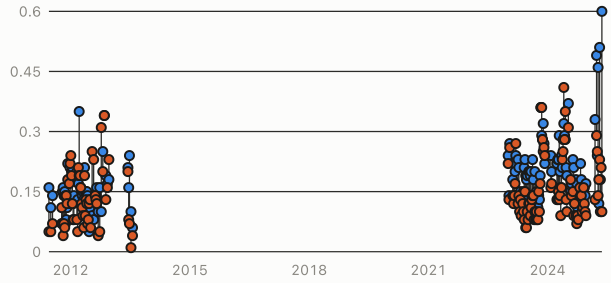
Botshol

grab sample (steekmonster) · 2018–2025 · outflow cleaner on 96% of days



Vinkeveen

pooled sample (verzamelmonster) · 2011–2025 · outflow cleaner on 78% of days



Each dot pair is one day; the gray line between them is the drop. 3 samples above 0.6 mg/l are drawn at the top edge (hover shows the real value).

Fair comparison: removal at the same inflow level

Cleaner incoming water is harder to clean further, so we compare the plants when similar amounts of phosphorus come in

Phosphorus coming in: 0.1-0.2 mg/l

Botshol (55 days) — 65% removed

Vinkeveen (84 days) — 31% removed

Phosphorus coming in: 0.2-0.4 mg/l

Botshol (67 days) — 76% removed

Vinkeveen (43 days) — 22% removed

Caveats: Botshol uses single grab samples and Vinkeveen pooled samples, which may explain part of the gap. Vinkeveen has no paired samples from 2014 to 2022. This shows what the plant does to the water passing through it, not yet the effect on the lake.

> → **Finding:** both plants work, but Botshol removes **71%** of phosphorus and Vinkeveen only **30%**, even when the same amount comes in. Something Botshol does could make Vinkeveen twice as effective.

5c · Early warning: forecasting algae blooms about 4 weeks ahead (Model 2)

Model 2 forecasts the algae level at a spot's **next** sample and warns for a bloom (more than 100 µg/l). It only uses what is already known on the day of the last sample. We compare it to the simplest guess: "same as last time".

More blooms caught

+16%

527 vs 453 of 997 blooms, same number of warnings

Fewer false alarms

−12%

548 vs 622

Algae level explained (R^2)

0.55

"same as last time": 0.24

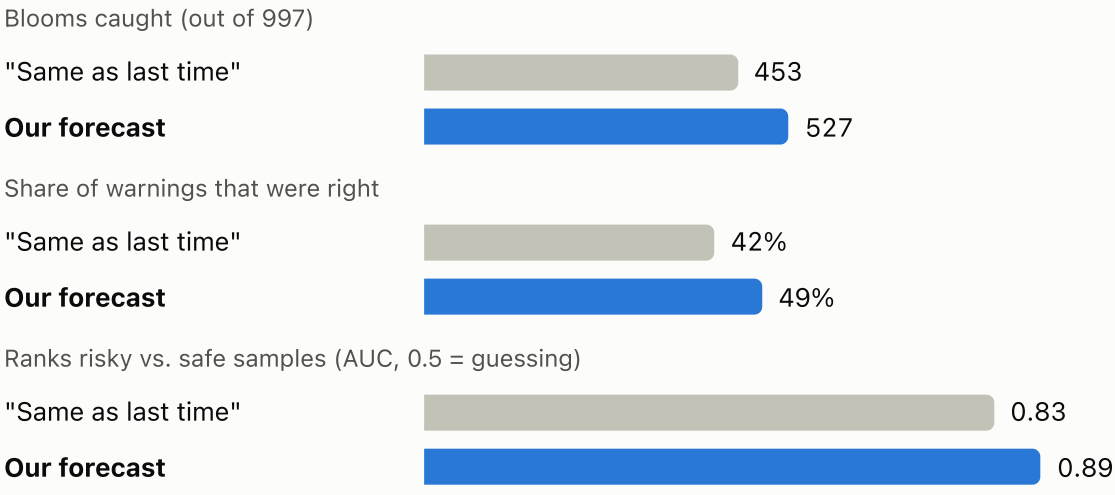
Forecast horizon

~4 weeks

median time to next sample · 11,001 test forecasts

Forecast vs. "same as last time" (test years 2020–2025, never seen in training)

Both methods raise the same number of warnings (1,075), so the comparison is fair.

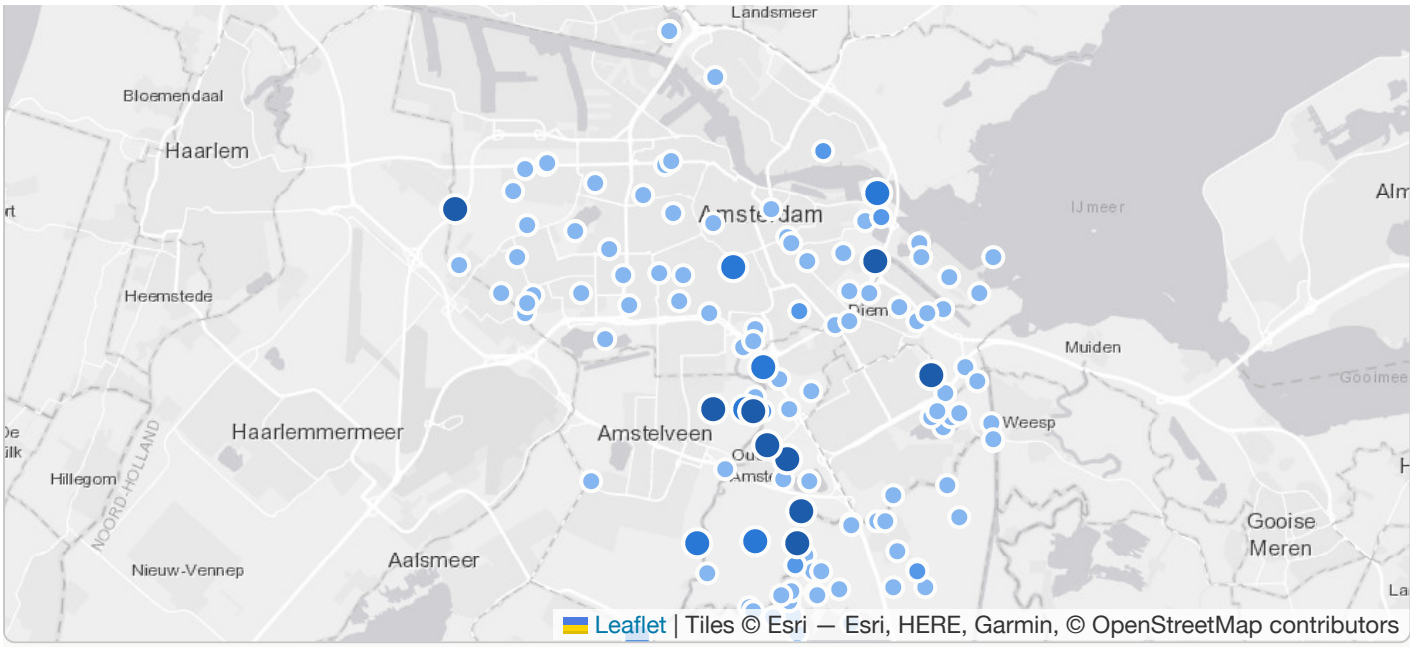


Bloom risk for June 2025, per spot

How Waternet would use it: each spot's latest sample (most from 2025-05) goes in, and out comes the chance of a bloom about 4 weeks later. Darker = higher risk, so sample or act there first.

< 10% (150 spots) · 10–25% (15 spots) · 25–50% (10 spots) · 50–75% (10 spots) · > 75% (1 spot)





Limits: it still misses about half of blooms and about half of its warnings are false, so it shows where to look first; it's not a guarantee. Adding weather barely improved it (R^2 0.55 → 0.57).

> → **Finding:** about 4 weeks ahead, with data Waternet already has, the forecast catches **16% more blooms** than "same as last time", with **12% fewer false alarms**.

5d · Climate stress test: what if summers get warmer?

We ran Model 1 on recent summer samples with air and water 1 °C and 2 °C warmer, keeping everything else the same.

Today (summers 2020–2024)

22.3 µg/l

average expected algae

+1 °C warmer

+4%

23.1 µg/l average expected algae

+2 °C warmer

+7%

23.9 µg/l average expected algae

Likely an underestimate: the model never saw water warmer than 24 °C, and 8% of the +2 °C samples go beyond that, so it treats them like the warmest it knows. It's a what-if based on patterns in the data, not a guarantee. Based on 2,769 summer samples at 118 spots.

> → **Finding:** 2 °C warmer summers mean **at least ~7% more algae**, which is likely an underestimate. Climate change works against the clean-up.

6 · Conclusions and suggested actions

What we found	What we suggest	How sure are we?
Cleaning up stalled around 2010. The approach that worked before (cutting nutrients from sewage and farms) has reached its limit.	Add place-specific measures on top of the current approach (below).	High measured directly
Problems stack up in the same places (110 of 235 spots break 3+ limits; clay ditches have the most stacked problems: 66% break 3 or more limits, versus 5% for lakes.)	Treat these as priority areas: one combined plan per area, instead of one measure per substance.	High measured directly
74 of 102 algae hotspots are driven by the type of place (peat, seepage, ditch bottoms releasing nutrients).	For peat areas: **keep peat wet** (stable, higher water levels) so it releases fewer nutrients. **Dredge** nutrient-rich mud from ditch bottoms. **Let in less sulphate-rich outside water** in summer. Bonus: wet peat also slows land subsidence.	Medium a strong pattern, tested on unseen spots, but not proof of cause
Botshol's plant removes 71% of phosphorus, Vinkeveen's 30%.	Find out what Botshol does differently and apply it at Vinkeveen. Then measure whether the lakes themselves get cleaner.	High for the difference; Unknown why
Blooms can be forecast ~4 weeks ahead better than "same as last time".	Use the monthly risk map to decide where to sample, inspect or warn swimmers first. Add sensors or satellite data to make it sharper.	Medium tested on 2020–2025, but misses half of blooms
Warmer summers add algae (at least ~7% at +2 °C).	Include climate in long-term targets: today's measures may not be enough in 2050.	Low–medium a model-based what-if

Tools: Python (pandas, scikit-learn), Leaflet maps, Claude Code as AI coding assistant. Data: Waternet, Waternet public pump register, KNMI. All analysis was done on the day of the hackathon.